Energy-Efficient Hybrid STAR-RIS via Sparse Active Selection and Gain Control

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***Abstract*—Hybrid active-passive simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RISs) can improve spectral efficiency (SE), but their energy efficiency (EE) is limited by amplifier noise and non-negligible hardware power. This letter studies an achievable EE-SE boundary with a multi-component power model under the energy-splitting (ES) protocol, including base station (BS) power-amplifier power, controller/bias power, switching overhead, and gain-dependent amplification power. The analysis yields an amplification ON-OFF criterion, a closed-form SE-oriented gain law, and an EE-oriented power-minimizing gain update, which are integrated with sparse active selection. Simulations under Nakagami-m fading show that the proposed selection and gain-control procedure improves the EE-SE tradeoff by activating only cost-effective elements, reducing BS transmit power at high-SE targets while avoiding excessive active-noise amplification and hardware overhead.**

***Index Terms*—STAR-RIS, active RIS, hybrid active-passive RIS, energy efficiency, spectral efficiency, Pareto optimization, amplifier noise.**

I. Introduction

Reconfigurable intelligent surfaces (RISs) can enhance wireless propagation through passive beam manipulation [1]–[3]. Simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RISs) extend this capability to two-side service, allowing one surface to support transmission-side and reflection-side users at the same time [4], [5]. This two-side operation is attractive for asymmetric coverage, but passive STAR-RIS still suffers from multiplicative path loss, especially when moderate or high spectral efficiency is required.

A direct way to compensate for this loss is to embed active elements into the surface. However, active amplification is not a free array gain: it injects amplifier noise and introduces hardware power consumption [6]–[10]. Measurement-driven RIS studies further show that controller, bias, and switching power should be modeled explicitly rather than absorbed into a single constant [13], [14]. These effects are particularly relevant for STAR-RIS because the two sides share aperture, power splitting, and the active-element budget.

Existing active and hybrid RIS/STAR-RIS studies have optimized transmit power, outage, or EE under different hardware assumptions [5]–[11], while measurement-driven studies model controller, unit-cell, and switching-related RIS power separately [13], [14]. However, the joint impact of amplifier noise, sparse activation, dynamic switching overhead, and side-wise rate allocation on the EE-SE tradeoff is still not captured under a common ES boundary-generation framework. This motivates an ES-focused design in which the active-noise and hardware-power terms remain explicit in both the gain analysis and the active-selection rule. The contributions are threefold. First, a multi-component EE model is formulated for hybrid active-passive STAR-RIS under ES, separating BS power-amplifier (PA) power, controller/bias consumption, switching overhead, gain-dependent amplification power, and active noise. Second, an analytical gain structure is derived, including an amplification ON-OFF criterion, a closed-form SE-oriented gain law, and an EE-oriented power-minimizing gain rule. Third, a finite-grid active-set selection and gain-control algorithm is developed to generate achievable EE-SE boundary points under binary activation, side assignment, gain update, and feasibility checking.

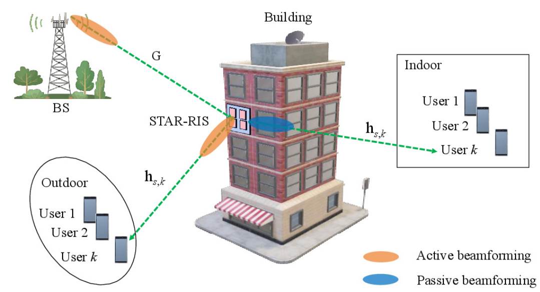

Fig. 1. Hybrid STAR-RIS-assisted downlink.

II. System Model

A. Hybrid STAR-RIS Downlink

Consider the downlink in Fig. 1, where a BS with M antennas serves K_t transmission-side users and K_r reflection-side users through an N-element STAR-RIS. The user-index sets are $\mathcal{K}_t = \{1, \ldots, K_t\}$ and $\mathcal{K}_r = \{1, \ldots, K_r\}$. Under the ES protocol [5], the nth STAR-RIS element is parameterized as

$$\theta_{n,s} = \sqrt{\rho_{n,s}}\beta_{n,s}e^{j\phi_{n,s}}, \qquad s \in \{t, r\}, \tag{1}$$

where $\rho_{n,s}$ is the power-splitting coefficient, $\beta_{n,s}$ is the amplification factor, and $\phi_{n,s}$ is the phase shift. The coefficients satisfy $\rho_{n,t} + \rho_{n,r} = 1$, $0 \leq \rho_{n,s} \leq 1$, and $1 \leq \beta_{n,s} \leq \beta_{\max}$. Passive elements use $\beta_{n,s} = 1$, whereas active elements use $\beta_{n,s} > 1$ on the side to which they are assigned. Let $\mathcal{A}_t$ and $\mathcal{A}_r$ denote the active-element sets for transmission and reflection. Each active element is assigned to at most one side. Hence $\mathcal{A}_t \cap \mathcal{A}_r = \emptyset$ and $\mathcal{A} = \mathcal{A}_t \cup \mathcal{A}_r$.

Denote by $\mathbf{G} \in \mathbb{C}^{N\times M}$ the BS-to-STAR-RIS channel, and by $\mathbf{h}_{s,k} \in \mathbb{C}^N$ the STAR-RIS-to-user channel on side s for

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Repository: https://github.com/hjhuang1989-svg/hybrid-star-ris.

user k. The received signal at user (s,k) is

$$y_{s,k} = \mathbf{h}_{s,k}^H \mathbf{\Theta}_s \mathbf{G} \sum_{q\in\{t,r\}} \sum_{j\in\mathcal{K}_q} \mathbf{w}_{q,j} x_{q,j} + \mathbf{h}_{s,k}^H \mathbf{n}_{a,s} + n_{s,k}, \quad (2)$$

where $\mathbf{\Theta}_s = \mathrm{diag}(\theta_{1,s},\ldots,\theta_{N,s})$, the summation index $q \in \{t,r\}$ enumerates the side of each transmitted stream, $\mathbf{w}_{q,j} \in \mathbb{C}^M$ is the BS beamformer for user (q,j), $x_{q,j}$ is the unit-power information symbol, and $n_{s,k} \sim \mathcal{CN}(0,\sigma_0^2)$ is thermal noise. The vector $\mathbf{n}_{a,s}$ represents the amplified noise contributed by active elements on side s. Following the active-RIS noise model in [7], [10], it is modeled as

$$\begin{aligned} &\mathbf{n}_{a,s} \sim \mathcal{CN}(\mathbf{0}, \mathbf{C}_{a,s}), \\ &\mathbf{C}_{a,s} = \mathrm{diag}(c_{1,s},\ldots,c_{N,s}). \end{aligned} \quad (3)$$

where $c_{n,s} = \rho_{n,s}\beta_{n,s}^2\sigma_a^2$ for $n \in \mathcal{A}_s$ and $c_{n,s} = 0$ otherwise, with σ_a^2 denoting the input-referred active-element noise variance. The SINR of user (s,k) is then

$$\gamma_{s,k} = \frac{\left|\mathbf{h}_{s,k}^H \mathbf{\Theta}_s \mathbf{G} \mathbf{w}_{s,k}\right|^2}{\sum_{\substack{q\in\{t,r\},\ j\in\mathcal{K}_q \\ (q,j)\neq(s,k)}} \left|\mathbf{h}_{s,k}^H \mathbf{\Theta}_s \mathbf{G} \mathbf{w}_{q,j}\right|^2 + \sigma_{a,s,k}^2 + \sigma_0^2}, \quad (4)$$

where $\sigma_{a,s,k}^2 = \mathbf{h}_{s,k}^H \mathbf{C}_{a,s} \mathbf{h}_{s,k}$ is the active-noise term seen at user (s,k).

B. Power-Consumption Model

Following measurement-based RIS power models [13], [14], the total consumed power is written as

$$\begin{aligned} P_{\mathrm{tot}} = &\frac{1}{\eta_{\mathrm{PA}}} \sum_{s\in\{t,r\}} \sum_{k\in\mathcal{K}_s} \|\mathbf{w}_{s,k}\|^2 + P_{\mathrm{ctrl}} + P_{\mathrm{bias}}|\mathcal{A}| \\ &+ f_u(E_{\mathrm{sw},0} + E_{\mathrm{sw},1}|\mathcal{A}|) \\ &+ \sum_{n\in\mathcal{A}_t} \mu_t(\beta_{n,t}^2 - 1) + \sum_{n\in\mathcal{A}_r} \mu_r(\beta_{n,r}^2 - 1), \end{aligned} \quad (5)$$

where η_{PA} is the BS PA efficiency, $|\mathcal{A}|$ is the number of active elements, P_{ctrl} is the controller power, P_{bias} is the per-active-element bias power, f_u is the update frequency, $E_{\mathrm{sw},0}$ and $E_{\mathrm{sw},1}$ are the base and per-active-element switching energies per update, and μ_t, μ_r are the gain-dependent amplifier-power coefficients. The four components in (5) are the PA-normalized BS transmit power, static controller/bias power, update-rate-weighted switching power, and gain-dependent amplification power.

C. Equivalent Scalar Model

To obtain a tractable EE-SE boundary, the full model is reduced to an equivalent scalar form using four approximations: residual multiuser leakage in (4) is neglected, common side-wise splitting and gain are used with $\rho_{n,s} = \rho_s$ and $\beta_{n,s} = \beta_s$ for $n \in \mathcal{A}_s$, dominant cascaded paths are phase-aligned, and the side target R_s is enforced by assigning the same target R_s/K_s to every user on side s, so that the K_s user targets sum to R_s.

Let $\mathbf{v}_{s,k}$ be a normalized post-zero-forcing (post-ZF) BS direction, and let $\mathbf{e}_n$ be the nth canonical basis vector in $\mathbb{R}^N$. The scalar BS-to-element gain is represented by the stream-averaged effective magnitude

$$|g_n| \triangleq \frac{1}{K_t + K_r} \sum_{q\in\{t,r\}} \sum_{k\in\mathcal{K}_q} \left|\mathbf{e}_n^T \mathbf{G} \mathbf{v}_{q,k}\right|.$$

With $h_{s,k,n}$ denoting the nth entry of $\mathbf{h}_{s,k}$, define

$$u_{s,k,n} = |g_n|\,|h_{s,k,n}|. \quad (6)$$

The side-wise effective coherent term for user (s,k) is then

$$\psi_{s,k}(\beta_s) = \sum_{n\in\mathcal{P}_s} \sqrt{\rho_s} u_{s,k,n} + \beta_s \sum_{n\in\mathcal{A}_s} \sqrt{\rho_s} u_{s,k,n}, \quad (7)$$

where $\mathcal{P}_s = \{1,\ldots,N\} \setminus \mathcal{A}_s$ contains passive elements and those active only for the opposite side. The corresponding active-noise term is

$$\zeta_{s,k}(\beta_s) = \sigma_0^2 + \rho_s\beta_s^2\sigma_a^2 \sum_{n\in\mathcal{A}_s} |h_{s,k,n}|^2. \quad (8)$$

Under this equal per-user target, the minimum side power required to meet the side target R_s is

$$p_s^{\min}(\beta_s) = \sum_{k\in\mathcal{K}_s} \frac{(2^{R_s/K_s} - 1)\zeta_{s,k}(\beta_s)}{\psi_{s,k}^2(\beta_s)}. \quad (9)$$

III. Problem Formulation

Define the sum spectral efficiency as

$$R_\Sigma = \sum_{s\in\{t,r\}} \sum_{k\in\mathcal{K}_s} \log_2(1+\gamma_{s,k}). \quad (10)$$

Throughout the letter, energy efficiency is measured by $\mathrm{EE} = R_\Sigma / P_{\mathrm{tot}}$. Let $\mathcal{X} \triangleq \{\mathbf{W}, \rho_t, \rho_r, \boldsymbol{\phi}, \beta_t, \beta_r, \mathcal{A}_t, \mathcal{A}_r\}$. The EE maximization problem is

$$\begin{aligned} \max_{\mathcal{X}} \quad & \frac{R_\Sigma}{P_{\mathrm{tot}}} \\ \text{s.t.} \quad & \log_2(1+\gamma_{s,k}) \geq \bar{R}_{s,k},\ \forall s,k, \\ & \sum_{s\in\{t,r\}} \sum_{k\in\mathcal{K}_s} \|\mathbf{w}_{s,k}\|^2 \leq P_{\max}^{\mathrm{BS}}, \\ & 1 \leq \beta_s \leq \beta_{\max}, \quad s\in\{t,r\}, \\ & \mathcal{A}_t \cap \mathcal{A}_r = \emptyset, \quad \mathcal{A} = \mathcal{A}_t \cup \mathcal{A}_r, \\ & \rho_t + \rho_r = 1, \quad 0 \leq \rho_s \leq 1, \quad s\in\{t,r\}, \\ & \rho_s\beta_s^2\sigma_a^2 \leq \Gamma_s, \quad s\in\{t,r\}. \end{aligned} \quad (11)$$

Here, Γ_s is an optional side-wise noise-injection or hardware-safety limit. Problem (11) is non-convex because the fractional objective couples beamforming, STAR-RIS coefficients, side-wise gains, and active-set selection. To trace an achievable EE-SE Pareto tradeoff, the ϵ-constraint scalarization, a standard multicriteria-optimization approach [15], is adopted

$$\begin{aligned} \mathcal{P}(\epsilon): \quad \min_{\mathcal{X}} \quad & P_{\mathrm{tot}} \\ \text{s.t.} \quad & R_\Sigma \geq \epsilon, \\ & \text{constraints of (11)}. \end{aligned} \quad (12)$$

In the proposed procedure, the quality-of-service (QoS) terms are instantiated as $\bar{R}_{s,k} = R_s/K_s$, with $R_t = \alpha\epsilon$ and $R_r = (1-\alpha)\epsilon$, so that the per-user constraints realize the side targets in (9). Because the procedure uses structured sparse activation and finite grids, the generated curve is an achievable Pareto-frontier approximation rather than a certified global optimum. The model structure is then exploited to derive practical design rules.

IV. Gain Structure

A. Amplification ON-OFF Criterion

Given a side-specific active set, the key question is whether the side-wise active gain should be switched on, i.e., whether increasing β_s above one improves the side-wise SINR. For convenience, introduce four side-wise worst-case coefficients: a_s is the passive coherent contribution, b_s is the active coherent contribution evaluated at unit gain, c_s is the receiver thermal-noise baseline, and d_s is the active-noise coefficient before multiplication by β_s^2. They are

$$\begin{aligned} a_s &= \min_k \sum_{n\in\mathcal{P}_s} \sqrt{\rho_s} u_{s,k,n}, \\ b_s &= \min_k \sum_{n\in\mathcal{A}_s} \sqrt{\rho_s} u_{s,k,n}, \\ c_s &= \sigma_0^2, \\ d_s &= \max_k \rho_s \sigma_a^2 \sum_{n\in\mathcal{A}_s} |h_{s,k,n}|^2. \end{aligned} \tag{13}$$

The minimum in a_s and b_s protects the weakest user in coherent gain, while the maximum in d_s protects the user most exposed to active noise. Substituting a_s, b_s, c_s, and d_s into (7) and (8), and dropping the positive per-user power factor $p_{s,k}$ that does not affect monotonicity with respect to β_s, gives the normalized worst-user SINR lower bound

$$\underline{\gamma}_s(\beta_s) = \frac{(a_s + b_s\beta_s)^2}{c_s + d_s\beta_s^2}. \tag{14}$$

The coherent term grows linearly with β_s, whereas the injected active noise grows quadratically. Differentiating (14) gives

$$\frac{\mathrm{d}\underline{\gamma}_s}{\mathrm{d}\beta_s} = \frac{2(a_s + b_s\beta_s)(b_s c_s - a_s d_s \beta_s)}{(c_s + d_s\beta_s^2)^2}. \tag{15}$$

Thus, at the passive-gain point $\beta_s = 1$, amplification is locally useful if

$$\left.\frac{\mathrm{d}\underline{\gamma}_s}{\mathrm{d}\beta_s}\right|_{\beta_s=1} > 0 \iff b_s c_s > a_s d_s. \tag{16}$$

Equation (16) gives the amplification ON-OFF criterion: the active coherent-gain aggregate b_s, weighted by the thermal-noise baseline c_s, must dominate the passive coherent-gain aggregate a_s weighted by the active-noise coefficient d_s. When this inequality holds, amplification is locally beneficial for the SINR lower bound. Otherwise, the corresponding STAR-RIS amplifier should remain inactive during operation. For EE-oriented operation, bias, switching, and amplification-power costs make the ON-OFF decision more restrictive.

Proposition 1. *For the positive-coefficient hybrid active-passive case with $a_s > 0$, $b_s > 0$, $c_s > 0$, and $d_s > 0$, $\underline{\gamma}_s(\beta_s)$ is unimodal on $[1, \beta_{\max}]$. The SE-oriented gain is*

$$\beta^\star_{s,\mathrm{SE}} = \left[\frac{b_s c_s}{a_s d_s}\right]_{[1,\beta_{\max}]}, \tag{17}$$

where $[x]_{[l,u]} = \min\{u, \max\{l, x\}\}$. The boundary cases follow from the same derivative behavior. If $b_s = 0$, set $\beta^\star_{s,\mathrm{SE}} = 1$. If $a_s = 0$ with $b_s > 0$, or if $d_s = 0$ with $b_s > 0$, $\underline{\gamma}_s(\beta_s)$ is nondecreasing and $\beta^\star_{s,\mathrm{SE}} = \beta_{\max}$. If $|\mathcal{A}_s| = 0$, set $\beta_s = 1$.

Proof: When all four coefficients are positive, the sign of (15) is determined only by $b_s c_s - a_s d_s \beta_s$, whose sign changes only once. Hence (14) is unimodal and the stationary point is $b_s c_s/(a_s d_s)$, clipped to $[1, \beta_{\max}]$. The boundary cases follow directly from (15): $b_s = 0$ makes the derivative non-positive, while $a_s = 0$ with $b_s > 0$ or $d_s = 0$ with $b_s > 0$ makes (14) nondecreasing. ∎

B. Power-Minimizing Gain

The gain in (17) is SE-oriented because it maximizes the normalized worst-user SINR lower bound and does not include the denominator of EE. In contrast, the ϵ-constraint problem in (12) minimizes total consumed power at a fixed SE target. Increasing β_s may reduce the BS transmit power required by (9), but it also increases active noise and the gain-dependent amplifier power in (5). Therefore, the SE-oriented gain and the EE-oriented power-minimizing gain generally differ. For a target side rate R_s, define the side-wise cost $\Psi_s(\beta_s)$ as the sum of the transmit-power term induced by (9) under the worst-user coefficients and the gain-dependent active-amplifier power:

$$\Psi_s(\beta_s) = \frac{K_s(2^{R_s/K_s} - 1)(c_s + d_s\beta_s^2)}{\eta_{\mathrm{PA}}(a_s + b_s\beta_s)^2} + \mu_s|\mathcal{A}_s|(\beta_s^2 - 1), \tag{18}$$

$$\Psi_s'(\beta_s) = \frac{2K_s(2^{R_s/K_s} - 1)(a_s d_s \beta_s - b_s c_s)}{\eta_{\mathrm{PA}}(a_s + b_s\beta_s)^3} + 2\mu_s|\mathcal{A}_s|\beta_s, \tag{19}$$

$$\begin{aligned} \Psi_s''(\beta_s) = {}& \frac{2K_s(2^{R_s/K_s} - 1)}{\eta_{\mathrm{PA}}(a_s + b_s\beta_s)^4}\Big(a_s^2 d_s - 2a_s b_s d_s \beta_s + 3b_s^2 c_s\Big) \\ & + 2\mu_s|\mathcal{A}_s|. \end{aligned} \tag{20}$$

Proposition 2. *Assume the hybrid active-passive case $a_s > 0$, $b_s > 0$, and $d_s > 0$. Under the condition*

$$\beta_{\max} \le \frac{a_s^2 d_s + 3b_s^2 c_s}{2a_s b_s d_s}, \tag{21}$$

$\Psi_s(\beta_s)$ is strictly convex on $[1, \beta_{\max}]$. Consequently, the side-wise power-minimizing gain is unique and can be obtained by checking the two endpoints and the interior stationary point solving (19), if that point lies in $[1, \beta_{\max}]$.

Proof: Under (21), the bracketed term in (20) is non-negative on $[1, \beta_{\max}]$. Since $|\mathcal{A}_s| > 0$ and $\mu_s > 0$, the term $2\mu_s|\mathcal{A}_s|$ yields $\Psi_s''(\beta_s) > 0$ over the interval. Hence Ψ_s is strictly convex and has a unique minimizer. ∎

Condition (21) is sufficient but not necessary. When it is violated, $\Psi_s(\beta_s)$ can still be minimized efficiently on the bounded interval $[1, \beta_{\max}]$ by a one-dimensional line search. If no active element is assigned to side s, or if the selected elements produce no active coherent term ($b_s = 0$), the gain is set to $\beta_s = 1$ because increasing it cannot reduce the required power.

V. Sparse Active Selection and Gain Control

For a target sum-SE level ϵ in (12), this section constructs one feasible candidate of $\mathcal{P}(\epsilon)$. The target is split into side rates by $R_t = \alpha\epsilon$ and $R_r = (1-\alpha)\epsilon$, where $\alpha \in \mathcal{G}_\alpha$, while $L \in \mathcal{L}$ specifies the number of active elements. Use $x_{n,s} \in \{0, 1\}$, $s \in \{t, r\}$, with $x_{n,s} = 1$ iff element n is active on side s. The constraints $x_{n,t} + x_{n,r} \le 1$ and $\sum_{n=1}^N (x_{n,t} + x_{n,r}) = L$ impose side-exclusive activation with L active elements; $x_{n,t} = x_{n,r} = 0$ denotes a passive element, and $\mathcal{A}_s = \{n : x_{n,s} = 1\}$. Rather than solve the exact combinatorial program, Algorithm 1 constructs ranked binary candidates and evaluates them under the scalar model.

A. Selecting L Active Elements

For fixed (L, α), an element is useful if its coherent-gain contribution can reduce the BS power more than its added active-noise and hardware-power costs. Let

$$C_s = \frac{K_s(2^{R_s/K_s} - 1)}{\eta_{\mathrm{PA}}},$$

which collects the side-rate and PA-efficiency factors in the transmit-power term of (18). To derive a ranking score, introduce a temporary continuous insertion variable $\chi_{n,s} \in [0,1]$ only for local sensitivity analysis. The final decision variable remains the binary $x_{n,s}$ defined above.

If element n is tentatively inserted on side s, its worst-user coherent increment is $\Delta u_{n,s} = \sqrt{\rho_s}|g_n| \min_k |h_{s,k,n}|$, and its active-noise increment is approximated by $\Delta d_{n,s} \approx \rho_s \sigma_a^2 \mathbb{E}_k[|h_{s,k,n}|^2]$, where $\mathbb{E}_k[\cdot] \triangleq K_s^{-1} \sum_{k \in \mathcal{K}_s}(\cdot)$ is the empirical user average on side s. With $D_{n,s}(\chi_{n,s}) = a_s + b_s\beta_s + (\beta_s - 1)\chi_{n,s}\Delta u_{n,s}$, the insertion-relaxed side cost is approximated as

$$\begin{aligned}\Psi_s(\chi_{n,s}) \approx\, & C_s\big[c_s + \beta_s^2(d_s + \chi_{n,s}\Delta d_{n,s})\big] D_{n,s}^{-2}(\chi_{n,s}) \\ & + \mu_s(|\mathcal{A}_s| + \chi_{n,s})(\beta_s^2 - 1) \\ & + \chi_{n,s}(P_{\text{bias}} + f_u E_{\text{sw},1}).\end{aligned}$$

Differentiating at $\chi_{n,s} = 0$ gives the first-order benefit ratio

$$\tilde{\xi}_{n,s} \triangleq \frac{\kappa_{1,s}\Delta u_{n,s}}{\kappa_{2,s}\Delta d_{n,s} + P_{\text{bias}} + f_u E_{\text{sw},1} + \mu_s(\beta_s^2 - 1)},$$

where

$$\kappa_{1,s} = \frac{2C_s(c_s + d_s\beta_s^2)(\beta_s - 1)}{(a_s + b_s\beta_s)^3},$$

$$\kappa_{2,s} = \frac{C_s\beta_s^2}{(a_s + b_s\beta_s)^2}.$$

A value $\tilde{\xi}_{n,s} > 1$ indicates that the first-order coherent-gain saving dominates the active-noise, bias, switching, and amplifier-power penalties. For a non-iterative implementation, use side weights $\omega_t = \alpha$ and $\omega_r = 1 - \alpha$ and define the normalized score

$$\xi_{n,s} = \frac{\omega_s \Delta u_{n,s}}{\sigma_0^{-2}\Delta d_{n,s} + (P_{\max}^{\text{BS}})^{-1}(P_{\text{bias}} + f_u E_{\text{sw},1})}, \quad s \in \{t, r\}. \tag{22}$$

The numerator is the side-weighted coherent increment, while the denominator normalizes the active-noise increment and per-active-element hardware overhead.

B. Assigning Active Elements to Sides

For each (L, α), two candidate active-set pairs are constructed by the same rule: choose each element's preferred side, retain the L largest preferred scores, and split the retained indices according to their preferred sides. For the cost-aware pair, define $s_n^{\text{ca}} = \arg\max_{s\in\{t,r\}} \xi_{n,s}$ and $\xi_n^{\text{ca}} = \xi_{n,s_n^{\text{ca}}}$. Let $\mathcal{S}^{\text{ca}}$ contain the L largest values of ξ_n^{ca}, and set $\mathcal{A}_s^{\text{ca}} = \{n \in \mathcal{S}^{\text{ca}} : s_n^{\text{ca}} = s\}$ for $s \in \{t, r\}$. Thus, selection and side assignment are coupled by the same score.

The second pair is channel-strength based and is used as the strongest-greedy baseline in Sec. VI. It removes the denominator of (22) and uses $\pi_{n,s} = \omega_s \Delta u_{n,s}$. With $s_n^{\text{st}} = \arg\max_{s\in\{t,r\}} \pi_{n,s}$ and $\pi_n^{\text{st}} = \pi_{n,s_n^{\text{st}}}$, the L largest values of π_n^{st} form $\mathcal{S}^{\text{st}}$, and $\mathcal{A}_s^{\text{st}} = \{n \in \mathcal{S}^{\text{st}} : s_n^{\text{st}} = s\}$ for $s \in \{t, r\}$. Both candidate pairs are evaluated in Algorithm 1.

C. Updating (β_t, β_r)

For each candidate pair $(\mathcal{A}_t, \mathcal{A}_r)$, the coefficients (a_s, b_s, c_s, d_s) are computed from (13). The boundary cases in Proposition 1 are handled by setting $\beta_s = 1$ when $|\mathcal{A}_s| = 0$ or $b_s = 0$. Otherwise, the EE-oriented gain is updated by

$$\beta_s^\star = \arg \min_{1 \le \beta_s \le \beta_{\max}} \Psi_s(\beta_s),$$

where $\Psi_s(\beta_s)$ is given in (18). The updated gains determine the required side powers $p_t^{\min}(\beta_t^\star)$ and $p_r^{\min}(\beta_r^\star)$ through (9).

D. Feasibility Check and Boundary-Point Selection

After the gain update, each candidate is checked against the scalar-model version of (11): the BS transmit-power budget, the gain bounds, the non-overlapping active-set constraint, the ES splitting constraints, and the optional safety cap Γ_s. Among all feasible candidates for the current target ϵ, the returned boundary point is

$$\mathcal{B}^\star = \left(\mathcal{A}_t^\star, \mathcal{A}_r^\star, \beta_t^\star, \beta_r^\star, \alpha^\star, p_t^{\min,\star}, p_r^{\min,\star}, P_{\text{tot}}^\star, \text{EE}^\star\right),$$

where $P_{\text{tot}}^\star$ is the smallest feasible value of (5) and $\text{EE}^\star = \epsilon / P_{\text{tot}}^\star$. The complete procedure is summarized in Algorithm 1.

Algorithm 1 Sparse Active Selection and Gain Control

Initialize: $P_{\min} = +\infty$ and $\mathcal{B}^\star = \emptyset$.
for all $(L, \alpha) \in \mathcal{L} \times \mathcal{G}_\alpha$ **do**
  $R_t = \alpha\epsilon$ and $R_r = (1 - \alpha)\epsilon$.
  Form $(\mathcal{A}_t^{\text{ca}}, \mathcal{A}_r^{\text{ca}})$ using $\xi_{n,s}$ in (22).
  Form $(\mathcal{A}_t^{\text{st}}, \mathcal{A}_r^{\text{st}})$ using $\pi_{n,s}$.
  for all $(\mathcal{A}_t, \mathcal{A}_r) \in \{(\mathcal{A}_t^{\text{ca}}, \mathcal{A}_r^{\text{ca}}), (\mathcal{A}_t^{\text{st}}, \mathcal{A}_r^{\text{st}})\}$ **do**
    Compute (a_s, b_s, c_s, d_s) from (13) for $s \in \{t, r\}$.
    For each s, **if** $|\mathcal{A}_s| = 0$ or $b_s = 0$, set $\beta_s^\star = 1$.
    Otherwise
    set $\beta_s^\star = \arg\min_{1\le\beta_s\le\beta_{\max}} \Psi_s(\beta_s)$ for that side.
    Compute $p_t^{\min}$, $p_r^{\min}$, and P_{tot} from (9) and (5).
    If constraints of (11) hold and $P_{\text{tot}} < P_{\min}$, set $P_{\min} = P_{\text{tot}}$ and $\mathcal{B}^\star = (\mathcal{A}_t, \mathcal{A}_r, \beta_t^\star, \beta_r^\star, \alpha, p_t^{\min}, p_r^{\min}, P_{\text{tot}}, \epsilon/P_{\text{tot}})$.
  end for
end for
return $\mathcal{B}^\star$.

E. Complexity

Let $G = |\mathcal{G}_\alpha|$, $L_g = |\mathcal{L}|$, $K = K_t + K_r$, and Q_β be the bounded gain-search budget. For one SE target, the proposed finite-grid ranker requires $\mathcal{O}(L_g G(N \log N + NK + Q_\beta))$ operations for scoring, sorting, feasibility checking, and gain update; for E targets the cost scales linearly by E. The memory cost is dominated by the channel coefficients and element scores.

VI. Numerical Results

Simulations evaluate the EE-SE boundary, hardware sensitivity, power distribution, and candidate-search behavior. The baselines include passive ($L = 0$), active ($L = N$), fixed/optimized uniform, single-random sparse, and strongest greedy. Unless otherwise stated, $N = 24$, $K_t = K_r = 2$, and the Nakagami-m channels use $m_{\text{BR}} = m_{\text{RU}} = 1$ with $\Omega_{BR} = 0.20$, $\Omega_t = (0.060, 0.050)$, and $\Omega_r = (0.080, 0.060)$. We set $(\rho_t, \rho_r) = (0.55, 0.45)$, $\beta_{\max} = 5$, $P_{\max}^{\text{BS}} = 5$ W, $\sigma_0^2 = 6.25 \times 10^{-2}$, $\sigma_a^2 = 2.5 \times 10^{-3}$, and $\eta_{\text{PA}} = 0.40$. The grids are $\mathcal{G}_\alpha = \{0.20, 0.25, \ldots, 0.80\}$ and $\mathcal{L} = \{0, 3, \ldots, 24\}$. Motivated by measured RIS power models, $(P_{\text{ctrl}}, P_{\text{bias}}, f_u) = (0.10\text{ W}, 0.015\text{ W}, 50\text{ Hz})$, $(E_{\text{sw},0}, E_{\text{sw},1}) = (2.0, 1.2) \times 10^{-4}$ J, and $\mu_t = \mu_r = 2.5 \times 10^{-3}$ W [13], [14]. The channel and active-noise settings follow [5], [7], [8], [10]–[12], and each plotted Monte Carlo (MC) point averages 10000 independent realizations.

Fig. 2 shows the EE-SE boundary and the main power-allocation tradeoff. Passive STAR-RIS is competitive at low SE, but its BS-power requirement grows rapidly with R_Σ and becomes infeasible when it exceeds $P_{\max}^{\text{BS}}$. The unfilled passive markers at high R_Σ are averaged over feasible trials only, and their labels report the passive feasibility probability p_f^{pass}, which is 0.99 and 0.88 at $R_\Sigma = 18$ and 20 bit/s/Hz, respectively. The active benchmark lowers BS power but incurs substantial static, switching, and amplification overhead. The proposed and strongest-greedy curves are close in this setup because their selected active sets often overlap, while the proposed ranker keeps the finite-grid search cost-aware by penalizing active noise and hardware costs.

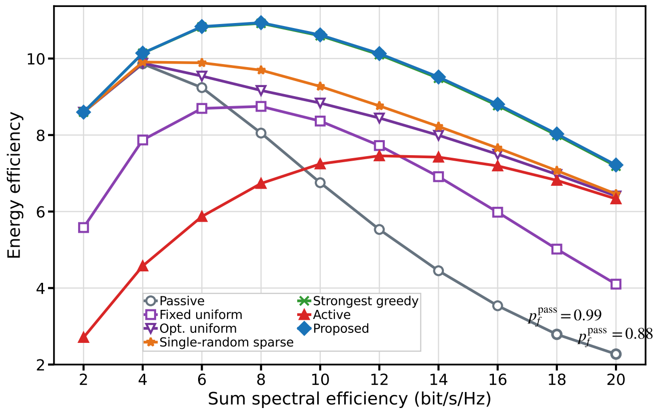


Fig. 2. Comparison of achievable EE-SE boundaries.

Fig. 3 shows hardware sensitivity at $R_\Sigma = 6$, 9, 12, 15, and 18 bit/s/Hz. A larger gain is not always beneficial: increasing σ_a suppresses the selected gain because amplifier noise scales with β_s^2 and raises the required power, as shown in Fig. 3(a). Increasing f_u similarly increases switching power and reduces EE, as shown in Fig. 3(b).

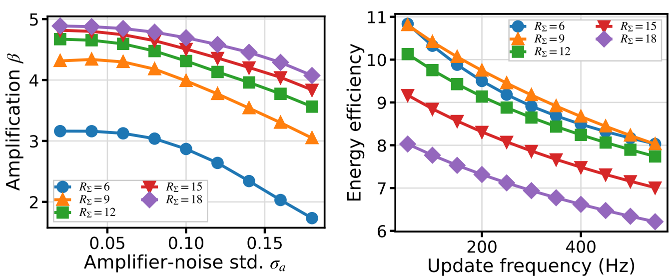
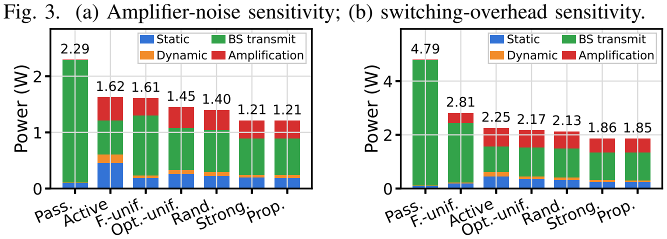


Fig. 3. (a) Amplifier-noise sensitivity; (b) switching-overhead sensitivity.



(a) $R_\Sigma = 12$ bit/s/Hz (b) $R_\Sigma = 16$ bit/s/Hz

Fig. 4. Power distribution sorted by total power: (a) $R_\Sigma = 12$ bit/s/Hz; (b) $R_\Sigma = 16$ bit/s/Hz.

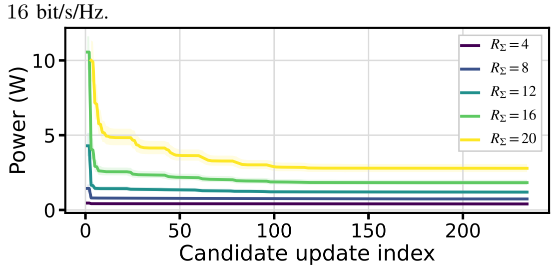


Fig. 5. Total power of the proposed finite candidate search.

Fig. 4 compares the power distribution at $R_\Sigma = 12$ and 16 bit/s/Hz: active elements reduce BS transmit power but add static, switching, and amplification power, so sparse activation provides the best balance. Fig. 5 shows the incumbent feasible total power during the finite candidate search; low-SE targets stabilize early, whereas high-SE targets need more (L, α) and side-assignment evaluations before the tradeoff settles.

VII. Conclusion

This letter studied the EE-SE behavior of hybrid active-passive STAR-RIS with sparse active selection and gain control. The results show that EE is maximized at a moderate active density: too few active elements require large BS transmit power, whereas too many amplify noise and accumulate hardware power. The cost-aware ranker avoids elements whose coherent gain is outweighed by active-noise and hardware penalties, and the search trace shows that demanding SE targets require more candidate evaluations before the power tradeoff stabilizes. Future work will consider full-dimensional beamforming and signal-dependent active-output constraints.

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